FINAL REPORT

My goal for this project is to track the relationship between my gym/workout level and my level of productivity. Gym/workout has become one of the most talked about topics in recent times. People are becoming more aware of the mental, physical and emotional benefits of working out. The projections on social media by gym enthusiasts have also influenced a good number of people to follow the path. Health and Fitness Association, in a recent survey reported a 31% increase in gym membership in USA. I am no exception to this because working out occupies an important space in my daily routine.

Gym has become, not just one of my go-to places when I need to reset, or shake off stress, but an integral part of my daily routine that sets my day on the right energetic trajectory. For some years now I start my day with workout, just after meditation and I get loaded up with a high achieving mindset and energy. Whereas when I don't train as much or I miss gym, there seems to be low level of energy towards work; a drop in alertness and getting my daily assignments done. Hitting the gym has become one of the dominant parts of my regular routine.

This project, therefore, seeks to ascertain if my workout is the major factor in my daily productivity or are there other superior or equal factors. Hitting the gym has become one of the dominant parts of my regular routine, based on this, I predicted that it has become the number one determinant of my productivity level; and has moved ahead of other variables like sleep and feeding

In exploring this, I got a diary and took a daily record of my workout, feeding, sleeping, meditation and how productive the day turns out to be. My data collection is as follows:

- Daily gym/workout: In tracking my gym session, I focused on the quality of the workout rather than time spent. I tracked the major compound workouts for strength and energy which are Bench-press and Squatting, with average weight of 80kg for as many reps I can go.
- Sleep: Is a vital factor on how someone responds to the events of the day and delivers. So, as part of my data, I tracked how much sleep I got (in minutes)
- Feeding: is another vital factor in productivity. In African parlance it is said that “a hungry man is an angry man” and so cannot be productive. I took records of the number of times I ate each day.
- Meditation: meditation is one act that lifts the mood and sets the brain and mind ready for the day. For this project, I recorded my daily meditation in minutes.
- Level of productivity: I calculated my daily productivity by the number of tasks I set to achieve each day and the number I successfully completed

Through this data collection, I was able to track the relationship between these variables and how they contribute to my daily productivity

Finally, my hypothesis says**: As my gym/workout level decreases, the level of my productivity decreases.**

DATASET DESCRIPTION AND RATIONALE

My dataset consists of daily personal observations collected between January and March 2026. Each daily record includes measurements of sleep duration, meditation duration, number of meals, gym/workout level, and productivity level. The goal of collecting this data is to ascertain how my gym/workout level affects my level of productivity. It further assesses the influence of other listed daily activities on my daily productivity.

Each observation is recorded as a single day, allowing analysis of correlations between gym, sleep, meditation, feeding and productivity level.

- Date: The day the observation was recorded.
- Gym Activity: Number of workout repetitions performed that day.
- Meals: Number of meals consumed that day.
- Meditation: Minutes spent meditating.
- Sleep Duration: Categorized as low, medium, or high.
- Productivity: Self-assessed productivity score ranging from 1 to 5.

The rationale for selecting these data variables is grounded in the hypothesis that increased physical activity leads to improved productivity. Gym/workout duration was chosen as the independent variable because it represents the behavior being tested. Productivity level was selected as the dependent variable because it reflects the outcome expected to change in response to workout activity.

Sleep duration, meditation and meal frequency were also collected as supporting variables because they are common factors that influence energy, focus, and cognitive performance. Including these variables allows for a more accurate interpretation of results by helping distinguish whether changes in productivity are associated with exercise specifically or influenced by other lifestyle factors. Collecting data across multiple days enables pattern observation and comparison over time, which is necessary to evaluate trends and relationships between variables.

DATA STRUCTURES

[{

date: 2026-01-23,
Sleep: 400mins, Meditation: 15mins, Meals: 2, Gym: { weightKg:85,reps:
210 }, Productivity: { tasksCompleted: 3, tasksSet: 4, Remark 3.75 }},
{
date: 2026-01-24,

Sleep: 345mins, Meditation: 16mins, Meals: 2, Gym: { weightKg:85, reps: 180 }, Productivity: { tasksCompleted: 2, tasksSet: 4, Remark:2.5 } },
{
date: 2026-01-26,
Sleep: 305mins, Meditation: 15mins, Meals: 1, Gym: { weightKg:0, reps: 0 }, Productivity: { tasksCompleted: 1, tasksSet: 3, Remark: 1.67 } },
{
date: 2026-01-27,
Sleep: 330mins, Meditation: 14mins, Meals: 2, Gym: { weightKg: 87, reps: 200 }, Productivity: { tasksCompleted: 2, tasksSet: 3, Remark: 3.33 } },
{
date: 2026-01-28,
Sleep: 420mins, Meditation: 15mins, Meals: 3, Gym: { weightKg:90, reps: 240 } Productivity: { tasksCompleted: 3, tasksSet: 3, Remark: 5 } },
{
date: 2026-01-29,
Sleep: 240mins, Meditation: 18mins, Meals: 2, Gym: { weightKg: 0, reps: 0 }, Productivity: { tasksCompleted: 3, tasksSet: 4, Remark: 3.75 } },
{
date: 2026-01-31,
Sleep: 420mins, Meditation: 15mins, Meals: 2, Gym: {weightKg:82, reps: 80}, Productivity: {tasksCompleted: 2, tasksSet: 2, Remark: 5} },
{
date: 2026-02-02,
Sleep: 360mins, Meditation: 14mins, Meals: 1, Gym: {weightKg:0, reps: 0 },Productivity: {tasksCompleted: 3, tasksSet: 3, Remark: 5 }},
{
date: 2026-02-03,
Sleep: 300mins, Meditation: 15mins, Meals: 2, Gym: {weightKg: 0, reps: 0}, Productivity: {tasksCompleted: 1, tasksSet: 4, Remark: 1.25} },
{
date: 2026-02-04,
Sleep: 340mins, Meditation: 15mins, Meals: 1, Gym: {weightKg: 80, reps: 70}, Productivity: {tasksCompleted: 3, tasksSet: 4, Remark: 3.75} },
{
date: 2026-02-05,
Sleep: 300mins, Meditation: 10mins, Meals: 2, Gym: {weightKg: 85, reps: 160}, Productivity: {tasksCompleted: 1, tasksSet: 2, Remark: 2.5 } },
{
date: 2026-02-06,
Sleep: 300mins, Meditation: 15mins, Meals: 2, Gym: {weightKg: 90, reps: 110}, Productivity: {tasksCompleted: 2, tasksSet: 3, Remark: 3.33 } },
{

date: 2026-02-07,
Sleep: 360mins, Meditation: 15mins, Meals: 2, Gym: {weightKg: 85, reps: 120}, Productivity: {tasksCompleted: 3, tasksSet: 3, Remark: 5 } },
{
date: 2026-02-09,
Sleep: 480mins, Meditation: 14mins, Meals: 2, Gym: {weightKg: 85, reps: 100}, Productivity: { tasksCompleted: 3, tasksSet: 4, Remark: 3.75} },
{
date: 2026-02-14,
Sleep: 360mins, Meditation: 6mins, Meals: 3, gym: { weightKg: 0, reps: 0}, Productivity: {tasksCompleted: 2, tasksSet: 4, Remark: 2.5} },
{
date: 2026-02-16,
Sleep: 360mins, Meditation: 9mins, Meals: 2, Gym: { weightKg: 80, reps: 200}, Productivity: {tasksCompleted: 3, tasksSet: 4, Remark: 3.75 }},
{
date: 2026-02-23,
Sleep: 405mins, Meditation: 8mins, Meals: 2, Gym: { weightKg: 82, reps: 210}, Productivity: {tasksCompleted: 4, tasksSet: 4, Remark: 5}},
{
date: 2026-02-24,
Sleep: 390mins, Meditation: 12mins, Meals: 2, Gym: {weightKg: 78, reps: 210}, Productivity: {tasksCompleted: 5, tasksSet: 5, Remark: 5} },
{

date: 2026-02-25,
Sleep: 360mins, Meditation: 12mins, Meals: 1, Gym: {weightKg: 80, reps: 50}, Productivity: {tasksCompleted: 4, tasksSet: 4, Remark: 5} },
{
date: 2026-03-05,
Sleep: 390mins, Meditation: 10mins, Meals: 2, Gym: { weightKg: 90, reps: 200 }, Productivity: {tasksCompleted: 3, tasksSet: 3, Remark:5 }},
{
date: 2026-03-06,
Sleep: 405mins, Meditation: 8, Meals: 2, Gym: {weightKg: 90, reps: 200}, Productivity: {tasksCompleted: 3, tasksSet: 3, Remark: 5} },
{
date: 2026-03-07,
Sleep: 390mins, Meditation: 10mins, Meals: 2, Gym: { weightKg: 90, reps: 210 }, Productivity: {tasksCompleted: 3, tasksSet: 4, Remark: 3.75} },
{
date: 2026-03-08,
Sleep: 400mins, Meditation: 5mins, Meals: 1, Gym: { weightKg: 0, reps: 0 }, Productivity: { tasksCompleted: 2, tasksSet: 3, Remark: 3.75} },

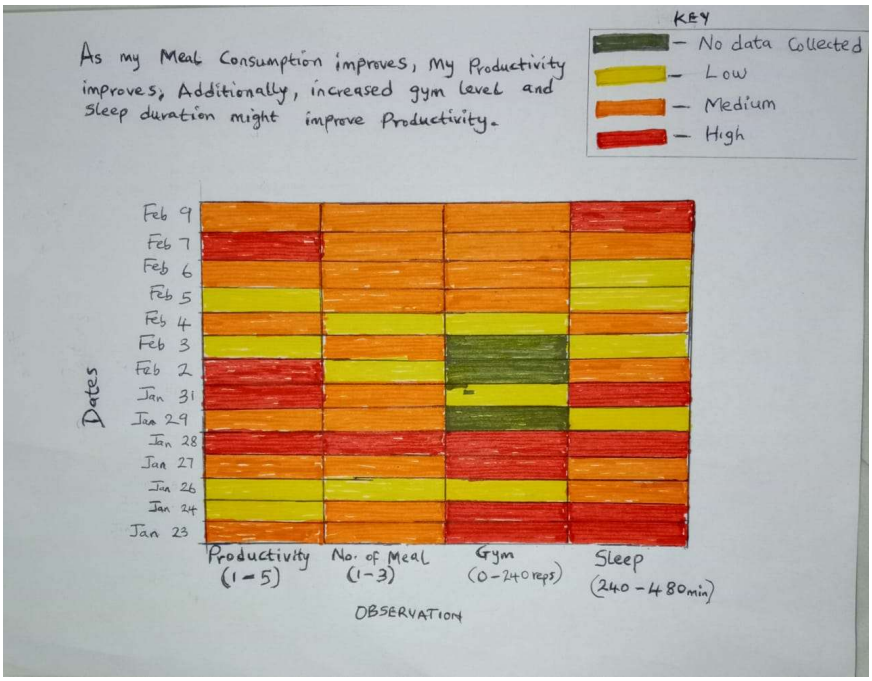
{
date: 2026-03-09,
Sleep: 300mins, Meditation: 8mins, Meals: 3, Gym: {weightKg: 92, reps: 250 }, Productivity: {tasksCompleted: 4, tasksSet: 4, Remark: 5} },
{
date: 2026-03-10,
Sleep: 420mins, Meditation: 10mins, Meals: 2, Gym: {weightKg: 90, reps: 150}, Productivity: {tasksCompleted: 3, tasksSet: 4, Remark: 3.75} },
{
date: 2026-03-11,
Sleep: 400mins, Meditation: 8mins, Meals: 2, Gym: {weightKg: 90, reps: 240}, Productivity: {tasksCompleted: 3, tasksSet: 3, Remark: 5 } },
{
date: 2026-03-12,
Sleep: 420mins, Meditation: 10mins, Meals: 2, Gym: {weightKg: 90, reps: 240}, Productivity: {tasksCompleted: 3, tasksSet: 4, Remark: 3.75} }]

VISUALIZATION RATIONALE

The visualization uses a scatterplot format to represent relationships between the variables (gym, sleep, feeding and productivity). Each rectangle represents one day of observations. Multiple visual encodings are used to display different aspects of the dataset simultaneously. I used scatterplot for my visualization because it is clear, straightforward and simple to read.

Scatterplot clearly defines my data and the relationship they share and gives a better focused result for my hypothesis. This is unlike heatmap visualization which, although portrays the relationship my variables share, does not have a focused picturesque result for my hypothesis as scatterplot.

Below is the sketch of the heatmap:



SELF-ANALYSIS

The hypothesis I explored in my final visualization is: As my workout increases, my productivity increases. I want to know how much influence gym has on my daily productivity. I further assessed the influence of other daily activities like sleep and feeding on my level of productivity.

From my data collection, scatterplot visualization suggests that higher workout activity has an impact on my level of productivity, but the influence is not absolute. There are days that the result from workout did not justify the level of productivity. Other variables like sleep and feeding prove to also have impact because days with sufficient sleep and regular meals appear to correspond with strong productivity outcome.

The result rather justifies that there is no single variable that absolutely controls my productivity level. Some exceptions exist in the dataset, indicating that productivity is influenced by multiple factors especially when all the variables are high.

I am aware that good sleep duration and feeding is very important for a person's efficiency, both in gym and other daily assignments, however I am surprised that workout is not an absolute determinant for my daily productivity level. This is because of how dedicated I have been to workout for about a decade and how it practically feels me up each day and sets me on high efficiency state.

It is interesting to see how habit tracking helps us understand how important every activity is and how improving on all fronts yields better results. Before now, I didn’t record the data of my daily activities. I only assessed them based on how I feel about each of them.

WORK CITED

Fitness & Health Association. *https://www.healthandfitness.org*. 30 October 2025. Google. 20 January 2026.